Chimwemwe Chinkuyu

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Education

Princeton University, Princeton, NJ — Class of 2026

B.S.E., Mechanical and Aerospace Engineering — Minor in Computer Science

Relevant Coursework: Eng. Design, Mech. Design, Robotics & Intelligent Systems, Heat Transfer, Data Structures and Algorithms

Technical Skills

CAD/CAM: Fusion 360, SolidWorks, PTC Creo, Onshape | **Software:** Python, MATLAB, Java, C, Simulink, HTML

Fabrication and Analysis: FEA, 3D-printing (FDM, SLA, SLS), CNC milling, manual machining, lathing, laser cutting, basic GD&T

Robotics: Rhino and Grasshopper, ROS, UR3 Collaborative Robot programming, UR-RTDE

Internship Experience

Early Charm Ventures — Baltimore, MD

Engineering Intern — Jun. 2025-Aug. 2025

- Programmed and operated a MillRight CNC to machine 324 light fixture slats, developing Fusion 360 CAM toolpaths, reducing cycle time by up to 15%, and troubleshooted gantry and limit switch failures through manufacturer coordination.
- Supported R&D of high-temperature-resistant carbide ceramic 3D-printing by preparing powder blends, running Selective Laser Reaction Sintering (SLRS) experiments, and characterizing ceramic microstructures via SEM and XRD.
- Advanced development of an electromechanical patient privacy screen, including Fusion 360 CAD modeling, FDM 3D-printing of revised components, and troubleshooting an Arduino-based accelerometer alert circuit.

Ventura Medical Technologies — Barcelona, Spain

Engineering Intern — May 2023-Jul. 2023

- Enhanced Python/Jupyter tools to analyze DICOM files characterizing Pectus Excavatum, generating surgeon-specific implant placement reports; developed Multi-Planar Reconstruction (MPR) and multi-location sternal density scan modules.
- Analyzed 20+ CT scans in RadiAnt, producing patient-specific sternal measurement reports and surgical recommendations for Pectus Up implant placement.
- Researched and proposed redesign recommendations for the Lantitaps (ear plug removal) device prototype, including higher-voltage batteries and fluid pumps with greater flow rate and pressure to meet target specifications.
- Supported regulatory and commercial operations by maintaining the Pectus Up patient database, coordinating with distributors for surgery data, and developing patient-facing brochure content for the Pectus Up surgical kit.

Early Charm Ventures — Baltimore, MD

Engineering Intern — Jun. 2022-Aug. 2022

- Ran and troubleshooted Prusa Mini +, Formlabs 3, 3L, and Fuse 1 3D printers and failed prints.
- Collaborated with engineers and scientists to analyze and optimize customer .stl files for printability.
- Post-processed hundreds of FDM, FFF, SLA, and SLS parts of a variety of materials to customer satisfaction.
- Contributed to the optimization of fast-paced 3D-printing startup workflow as it increased production volume.

Select Projects

Senior Thesis — Autonomous Robotic Assembly System — Sep. 2025-Apr. 2026

- Designed and validated an autonomous, open-loop robotic assembly system using a UR3 arm; integrated a parametric toolpath generator (Python/Grasshopper), Cartesian motion planner (COMPAS FAB/ROS/MoveIt!), and pneumatic gripper controller.
- Designed and 3D-printed custom hardware in Fusion 360; achieved 100% pick accuracy and multi-layer assembly success; diagnosed and resolved 7 failure modes across 4 development phases.

Design and Manufacturing of a Search and Rescue Robot — Oct. 2025-Dec. 2025

- Collaborated with 10 classmates to design and build a 120-lb marsupial search-and-rescue robot with closed-loop navigation via light and ultrasonic sensing for autonomous terrain traversal and sub-robot deployment.
- Co-led design and fabrication of a 100-lb low-carbon steel chassis and 60:1 belt-driven drivetrain; created Creo Parametric CAD models and GD&T engineering drawings, performed FBD and FEA to verify structural margins, and machined and MIG-welded frame members.

Design and Manufacturing of a Load-Bearing Airfoil — Mar. 2023-May. 2023

- Collaborated with 8 classmates to design a 25.5-inch aluminum I-beam airfoil; led structural design and FEA of the core I-beam in PTC Creo, validating 97 lbf load capacity at 3.75 in. tip deflection, and created GD&T fabrication drawings.
- Contributed to manufacturing, including CNC milling of the I-beam and FDM 3D printing of structural ribs; airfoil exceeded design load, failing at 120 lbf via beam web torsion.